

Module: 2, Random Variable

(Two dimensional random variable)

Two dimensional random Variables

Our study of random variables and their probability distributions in the preceding sections was restricted to one-dimensional sample spaces, in that we recorded outcomes of an experiment as values assumed by a single random variable.

There will be situations, however, where we may find it desirable to record the simultaneous outcomes of several random variables.

For example, if an 18-wheeler is to have its tires serviced and X represents the number of miles these tires have been driven and Y represents the number of tires that need to be replaced, then $p(30000, 5)$ is the probability that the tires are used over 30,000 miles and the truck needs 5 new tires.

Joint probability distribution

The function $p(x, y)$ is a joint probability distribution or probability mass function of the discrete random variables X and Y if

$$p(x, y) \geq 0 \text{ and } \sum_{x \in X} \sum_{y \in Y} p(x, y) = 1,$$

where $p(x, y) = P(X = x, Y = y)$.

Problem 1

Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If X is the number of blue pens selected and Y is the number of red pens selected, find

- 1). the joint probability function $p(x, y)$*
- 2). $P(X + Y \leq 1)$.*

Solution

- The possible pairs of values (x, y) are $(0, 0), (0, 1), (1, 0), (1, 1), (0, 2),$ and $(2, 0)$.
- Now, $p(0, 1)$, represents the probability that a red and a green pen are selected. The total number of equally likely ways of selecting any 2 pens from the 8 is $8C_2 = 28$
- The number of ways of selecting 1 red from 2 red pens and 1 green from 3 green pens is $2C_1 \times 3C_1 = 6$.
- Hence, $p(0, 1) = \frac{6}{28} = \frac{3}{14}$.
- Similar calculations yield the probabilities for the other cases, which are presented in following table.

Solution Cont...

	$p(x, y)$	x			Row Totals
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

$$\text{For (2), } P(X + Y \leq 1) = p(0, 0) + p(0, 1) + p(1, 0)$$

$$= \frac{3}{28} + \frac{3}{14} + \frac{9}{28}$$

$$= \frac{9}{14}$$

Joint probability density function

The function $f(x, y)$ is a joint probability density function of the continuous random variables X and Y if

$$f(x, y) \geq 0 \text{ and } \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

Problem 2

A privately owned business operates both a drive-in facility and a walk-in facility. On a randomly selected day, let X and Y , respectively, be the proportions of the time that the drive-in and the walk-in facilities are in use, and suppose that the joint density function of these random variables is

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y), & \text{if } 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- 1). Verify $f(x, y)$ is a probability density function
- 2). Find $P(0 < X < \frac{1}{2}, \frac{1}{4} < Y < \frac{1}{2})$

Solution

For(1), $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1.$

$$\begin{aligned}
 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy &= \int_0^1 \int_0^1 \frac{2}{5} (2x + 3y) dx dy \\
 &= \int_0^1 \left[\frac{2x^2}{5} + \frac{6xy}{5} \right]_{x=0}^{x=1} dy \\
 &= \int_0^1 \left(\frac{2}{5} + \frac{6y}{5} \right) dy \\
 &= \left[\frac{2y}{5} + \frac{3y^2}{5} \right]_{y=0}^{y=1} \\
 &= \frac{2}{5} + \frac{3}{5} \\
 &= 1.
 \end{aligned}$$

Solution Cont...

For(2), $P\left(0 < X < \frac{1}{2}, \frac{1}{4} < Y < \frac{1}{2}\right)$

$$\begin{aligned}
 P\left(0 < X < \frac{1}{2}, \frac{1}{4} < Y < \frac{1}{2}\right) &= \int_{\frac{1}{4}}^{\frac{1}{2}} \int_0^{\frac{1}{2}} \frac{2}{5} (2x + 3y) dx dy \\
 &= \int_{\frac{1}{4}}^{\frac{1}{2}} \left[\frac{2x^2}{5} + \frac{6xy}{5} \right]_{x=0}^{x=\frac{1}{2}} dy \\
 &= \int_{\frac{1}{4}}^{\frac{1}{2}} \left(\frac{1}{10} + \frac{3y}{5} \right) dy \\
 &= \left[\frac{y}{10} + \frac{3y^2}{10} \right]_{y=\frac{1}{4}}^{y=\frac{1}{2}} \\
 &= \frac{1}{10} \left[\left(\frac{1}{2} + \frac{3}{4} \right) - \left(\frac{1}{4} + \frac{3}{16} \right) \right] = \frac{13}{160}.
 \end{aligned}$$

The marginal distributions

The marginal distributions of X alone and Y alone are

$$P_X(x) = \sum_{y \in Y} p(x, y) \text{ and } P_Y(y) = \sum_{x \in X} p(x, y), \text{ if } (X, Y) \text{ discrete, and}$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy \text{ and } f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx, \text{ if } (X, Y) \text{ continuous case}$$

The conditional distributions for discrete case

Let X and Y be two discrete random variables. The conditional distribution of the random variable Y given that $X = x$ is

$$P(Y/X) = \frac{p(x, y)}{P_X(x)}, \text{ provided } P_X(x) > 0$$

Similarly, the conditional distribution of X given that $Y = y$ is

$$P(X/Y) = \frac{p(x, y)}{P_Y(y)}, \text{ provided } P_Y(y) > 0.$$

The conditional distributions for continuous case

Let X and Y be two continuous random variables. The conditional distribution of the random variable Y given that $X = x$ is

$$f(y/x) = \frac{f(x, y)}{f_X(x)}, \text{ provided } f_X(x) > 0$$

Similarly, the conditional distribution of X given that $Y = y$ is

$$f(x/y) = \frac{f(x, y)}{f_Y(y)}, \text{ provided } f_Y(y) > 0.$$

Remark

If we wish to find the probability that the discrete random variable X falls between a and b when it is known that the discrete variable $Y = y$, we evaluate

$$P(a < X < b | Y = y) = \sum_{a < x < b} P(X|Y),$$

where the summation extends over all values of X between a and b .

When X and Y are continuous, we evaluate

$$P(a < X < b | Y = y) = \int_a^b f(x|y) dx. \text{ Similarly,}$$

$$P(a < Y < b | X = x) = \sum_{a < y < b} P(Y|X), \text{ for } (X, Y) \text{ is discrete case}$$

$$P(a < Y < b | X = x) = \int_a^b f(y|x) dy, \text{ for } (X, Y) \text{ is continuous case}$$

Problem 3

The joint probability mass function of two dimensional random variables (X, Y) is

	$p(x, y)$	x		
		0	1	2
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0
	2	$\frac{1}{28}$	0	0

- 1 Find the marginal distributions functions of X and Y
- 2 Find the conditional distribution function of X given $Y = 1$.

Solution

	$p(x, y)$	x			$P_Y(y)$
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
$P_X(x)$		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

The marginal distributions functions of X is $P_X(x) = \begin{cases} \frac{5}{14}, & x = 0 \\ \frac{15}{28}, & x = 1 \\ \frac{3}{28}, & x = 2. \end{cases}$

Solution Cont...

The marginal distributions functions of Y is $P_Y(y) = \begin{cases} \frac{15}{28}, & y = 0 \\ \frac{3}{7}, & y = 1 \\ \frac{1}{28}, & y = 2. \end{cases}$

For(2), To find the conditional distribution of X , given that $Y = 1$.

We need to find $P(X/Y) = \frac{p(x,y)}{P_Y(y)}$, provided $P_Y(y) > 0$., where $y = 1$.

That is, to find $P(X/1) = \frac{p(x,1)}{P_Y(1)}$

First we have find $P_Y(1) = \sum_{x=0}^{x=2} p(x,1) = \frac{3}{14} + \frac{3}{14} + 0 = \frac{3}{7}$.

Solution Cont...

Now, $P(X/1) = \frac{P(x,1)}{P_Y(1)} = \frac{7}{3}p(x,1)$, $x = 0, 1, 2$. Therefore,

$$P(0/1) = \frac{p(0,1)}{P_Y(1)} = \frac{7}{3}p(0,1) = \left(\frac{7}{3}\right) \left(\frac{3}{14}\right) = \frac{1}{2}$$

$$p(1/1) = \frac{p(1,1)}{P_Y(1)} = \frac{7}{3}p(1,1) = \left(\frac{7}{3}\right) \left(\frac{3}{14}\right) = \frac{1}{2}$$

$$p(2/1) = \frac{p(1,1)}{P_Y(1)} = \frac{7}{3}p(2,1) = \left(\frac{7}{3}\right) 0 = 0$$

Solution Cont...

The conditional distribution of X , given that $Y = 1$, is

$$P(X/1) = \begin{cases} \frac{1}{2}, & x = 0 \\ \frac{1}{2}, & x = 1 \\ 0, & x = 2. \end{cases}$$

Problem 4

The joint probability mass function of two dimensional random variables (X, Y) is

	$p(x, y)$	y			
		1	2	3	4
x	1	$\frac{4}{36}$	$\frac{2}{36}$	$\frac{5}{36}$	$\frac{1}{36}$
	2	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{1}{36}$	$\frac{2}{36}$
	3	$\frac{3}{36}$	$\frac{3}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
	4	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{5}{36}$

- 1 Find the marginal distributions functions of X and Y
- 2 Find the conditional distribution function of X given $Y = y$
- 3 Find the conditional distribution function of Y given $X = x$.

Solution

For(1),

	$p(x, y)$	y				$P_X(x)$
		1	2	3	4	
x	1	$\frac{4}{36}$	$\frac{2}{36}$	$\frac{5}{36}$	$\frac{1}{36}$	$\frac{12}{36}$
	2	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{7}{36}$
	3	$\frac{3}{36}$	$\frac{3}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{8}{36}$
	4	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{5}{36}$	$\frac{9}{36}$
$P_Y(y)$		$\frac{10}{36}$	$\frac{9}{36}$	$\frac{9}{36}$	$\frac{8}{36}$	1

Solution Cont...

The marginal distribution function of X is $P_X(x) = \begin{cases} \frac{12}{36}, & x = 1 \\ \frac{7}{36}, & x = 2 \\ \frac{8}{36}, & x = 3 \\ \frac{9}{36}, & x = 4 \end{cases}$

The marginal distribution function of Y is $P_Y(y) = \begin{cases} \frac{10}{36}, & y = 1 \\ \frac{9}{36}, & y = 2 \\ \frac{9}{36}, & y = 3 \\ \frac{8}{36}, & y = 4 \end{cases}$

Problem 5

The joint probability mass function of two dimensional random variables (X, Y) is

	$p(x, y)$	x		
		-1	0	1
y	0	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{15}$
	1	$\frac{3}{15}$	$\frac{2}{15}$	$\frac{1}{15}$
	2	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{2}{15}$

- ① Find the marginal distributions functions of X and Y
- ② Find the conditional distribution function of X given $Y = y$
- ③ Find the conditional distribution function of Y given $X = x$.

Problem 6

The joint probability mass function of two dimensional random variables (X, Y) is $p(x, y) = k(2x + 3y)$, $x = 0, 1, 2$ and $y = 1, 2, 3$.

- 1 Find k .
- 2 Find the marginal distributions functions of X and Y
- 3 Find the conditional distribution function of X given $Y = y$
- 4 Find the conditional distribution function of Y given $X = x$.
- 5 Find $P(X + Y > 3)$.

Problem 7

The joint probability mass function of two dimensional random variables (X, Y) is $p(x, y) = \frac{x+y}{21}, x = 1, 2, 3$ and $y = 1, 2$

- 1 Find the marginal distributions functions of X and Y
- 2 Find the conditional distribution function of X given $Y = y$
- 3 Find the conditional distribution function of Y given $X = x$.
- 4 Find $P(X + Y > 3)$. and $P(X < 3)$.

Problem 8

The joint density for the random variables (X, Y) , where X is the temperature change and Y is the proportion of the spectrum that shifts for a certain atomic particle, is

$$f(x, y) = \begin{cases} 10xy^2, & 0 < x < y < 1 \\ 0, & \text{elsewhere.} \end{cases}$$

- 1 Find $f_X(x)$, $f_Y(y)$ and $f(y|x)$.
- 2 Find the probability that the spectrum shifts more than half of the total observations, given that the temperature is increased by 0.25 unit.

Solution

For(1), The marginal density function of X is

$$\begin{aligned}
 f_X(x) &= \int_{-\infty}^{\infty} f(x, y) dy \\
 &= \int_x^1 10xy^2 dy \\
 &= \left[\frac{10xy^3}{3} \right]_{y=x}^{y=1} \\
 &= \frac{10x}{3} - \frac{10xx^3}{3} \\
 &= \frac{10x}{3} - \frac{10x^4}{3} \\
 &= \frac{10x(1 - x^3)}{3}, 0 < x < 1.
 \end{aligned}$$

Solution

The marginal density function of Y is

$$\begin{aligned}f_Y(y) &= \int_{-\infty}^{\infty} f(x, y) dx \\&= \int_0^y 10xy^2 dx \\&= \left[\frac{10x^2y^2}{2} \right]_{x=0}^{x=y} \\&= \frac{10y^4}{2} - 0 \\&= 5y^4, 0 < y < 1.\end{aligned}$$

Solution Cont...

The conditional distribution functions of Y and X is

$$\begin{aligned}
 f(y|x) &= \frac{f(x,y)}{f_X(x)} \\
 &= \frac{10xy^2}{\frac{10x(1-x^3)}{3}} \\
 &= \frac{3y^2}{1-x^3}, 0 < x < y < 1.
 \end{aligned}$$

$$\begin{aligned}
 f(x|y) &= \frac{f(x,y)}{f_Y(y)} \\
 &= \frac{10xy^2}{5y^4} = \frac{2x}{y^2}, 0 < x < y < 1.
 \end{aligned}$$

Solution Cont...

For(2),

$$\begin{aligned}P\left(Y > \frac{1}{2} | X = 0.25\right) &= \int_{\frac{1}{2}}^1 f(y|x = 0.25) dy \\&= \int_{\frac{1}{2}}^1 \frac{3y^2}{1 - (0.25)^3} dy \\&= \left[\frac{3y^3}{3(1 - 0.016)} \right]_{\frac{1}{2}}^1 \\&= \left[\frac{y^3}{0.98} \right]_{\frac{1}{2}}^1 \\&= \frac{1}{0.98} \left[1 - \frac{1}{8} \right] = \frac{7}{7.84} = 0.89\end{aligned}$$

Problem 9

Given the joint density function

$$f(x, y) = \begin{cases} \frac{x(1+3y^2)}{4}, & 0 \leq x \leq 2, 0 \leq y \leq 1 \\ 0, & \text{elsewhere.} \end{cases}$$

- 1 Find $f_x(x)$, $f_Y(y)$, $f(x|y)$ and $f(y|x)$.
- 2 Evaluate $P\left(\frac{1}{4} < X < \frac{1}{2} \mid Y = \frac{1}{3}\right)$.

Problem 10

The joint probability density function of two dimensional continuous random variables (X, Y) is

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y), & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{elsewhere.} \end{cases}$$

- 1 Find $f_x(x)$ and $f_Y(y)$.
- 2 Find $f(x|y)$ and $f(y|x)$.

Problem 11

Given the joint density function

$$f(x, y) = \begin{cases} xy^2 + \frac{x^2}{8}, & 0 \leq x \leq 2, 0 \leq y \leq 1 \\ 0, & \text{elsewhere.} \end{cases}$$

- 1 Compute $P(X > 1)$, $P(Y < \frac{1}{2})$, $P(X > 1 | Y < \frac{1}{2})$ and $P(Y < \frac{1}{2} | X > 1)$
- 2 Evaluate $P(X < Y)$ and $P(X + Y \leq 1)$.

Solution

$$\begin{aligned}P(X > 1) &= \int_0^1 \int_1^2 f(x, y) dx dy \\&= \int_0^1 \int_1^2 \left(xy^2 + \frac{x^2}{8} \right) dx dy \\&= \int_0^1 \left[\frac{x^2 y^2}{2} + \frac{x^3}{24} \right]_1^2 dy \\&= \int_0^1 \left[\left(2y^2 + \frac{1}{3} \right) - \left(\frac{y^2}{2} + \frac{1}{24} \right) \right] dy \\&= \frac{19}{24}\end{aligned}$$

Solution Cont...

$$\begin{aligned}P(Y < \frac{1}{2}) &= \int_0^{\frac{1}{2}} \int_0^2 f(x, y) dx dy \\&= \int_0^{\frac{1}{2}} \int_0^2 \left(xy^2 + \frac{x^2}{8} \right) dx dy \\&= \int_0^{\frac{1}{2}} \left[\frac{x^2 y^2}{2} + \frac{x^3}{24} \right]_0^2 dy \\&= \int_0^{\frac{1}{2}} \left[2y^2 + \frac{1}{3} \right] dy \\&= \left[\frac{2y^3}{3} + \frac{y}{3} \right]_0^{\frac{1}{2}} = \frac{1}{4}\end{aligned}$$

Solution Cont...

We know that $P(X > 1 | Y < \frac{1}{2}) = \frac{P(X > 1, Y < \frac{1}{2})}{P(Y < \frac{1}{2})}$.

$$\begin{aligned}
 P(X > 1, Y < \frac{1}{2}) &= \int_0^{\frac{1}{2}} \int_1^2 f(x, y) dx dy \\
 &= \int_0^{\frac{1}{2}} \int_1^2 \left(xy^2 + \frac{x^2}{8} \right) dx dy \\
 &= \int_0^{\frac{1}{2}} \left[\frac{x^2 y^2}{2} + \frac{x^3}{24} \right]_1^2 dy \\
 &= \int_0^{\frac{1}{2}} \left[\left(2y^2 + \frac{1}{3} \right) - \left(\frac{y^2}{2} + \frac{1}{24} \right) \right] dy \\
 &= \frac{5}{24}
 \end{aligned}$$

Solution Cont...

$$\begin{aligned}
 \text{Therefore, } P(X > 1 | Y < \tfrac{1}{2}) &= \frac{P(X > 1, Y < \tfrac{1}{2})}{P(Y < \tfrac{1}{2})} \\
 &= \frac{\frac{5}{24}}{\frac{1}{4}} \\
 &= \frac{5}{6}, \\
 \text{and } P(Y < \tfrac{1}{2} | X > 1) &= \frac{P(X > 1, Y < \tfrac{1}{2})}{P(X > 1)} \\
 &= \frac{\frac{5}{24}}{\frac{19}{24}} \\
 &= \frac{5}{19}.
 \end{aligned}$$

Solution Cont...

For(2),

$$\begin{aligned}
 P(X < Y) &= \int_0^1 \int_0^y f(x, y) dx dy \\
 &= \int_0^1 \int_0^y \left(xy^2 + \frac{x^2}{8} \right) dx dy \\
 &= \int_0^1 \left[\frac{x^2 y^2}{2} + \frac{x^3}{24} \right]_0^y dy \\
 &= \int_0^1 \left[\frac{y^4}{2} + \frac{y^3}{24} \right] dy \\
 &= \left[\frac{y^5}{10} + \frac{y^4}{96} \right]_0^1 = \frac{53}{480}
 \end{aligned}$$

Solution Cont...

$$\begin{aligned}P(X + Y \leq 1) &= \int_0^1 \int_0^{1-y} f(x, y) dx dy \\&= \int_0^1 \int_0^{1-y} \left(xy^2 + \frac{x^2}{8} \right) dx dy \\&= \int_0^1 \left[\frac{x^2 y^2}{2} + \frac{x^3}{24} \right]_0^{1-y} dy \\&= \int_0^1 \left[\frac{(1-y)^2 y^2}{2} + \frac{(1-y)^3}{24} \right] dy \\&= \frac{13}{480}\end{aligned}$$

Statistical Independence

Let X and Y be two random variables, discrete (continuous), with joint probability mass (density) function $p(x, y)$ ($f(x, y)$) and marginal distributions $P_X(x)$ and $P_Y(y)$, ($f_X(x)$ and $f_Y(y)$) respectively. The random variables X and Y are said to be statistically independent if and only if $p(x, y) = P_X(x)P_Y(y)$, for all (x, y) within their range.
 $(f(x, y) = f_X(x)f_Y(y), \text{ for all } (x, y) \text{ within their range.})$

Problem 12

Given the joint density function

$$f(x, y) = \begin{cases} \frac{x^3 y^3}{16}, & 0 \leq x \leq 2, 0 \leq y \leq 2. \\ 0, & \text{elsewhere.} \end{cases}$$

Show that the random variables X and Y are independent.

ANS: $f_X(x) = \frac{x^3}{4}$ and $f_Y(y) = \frac{y^3}{4}$.

Problem 13

The joint probability mass function of two dimensional random variables (X, Y) is

	$p(x, y)$	x		
		0	1	2
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0
	2	$\frac{1}{28}$	0	0

Show that the random variables X and Y are not independent.